Linear Programming Applications

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1.The SilComputers production case

SilComputers makes quarterly decisions about their product mix. While their full product line includes hundreds of products, we will consider a simpler problem with just two products: notebook computers and desktop computers. SilComputers would like to know how many of each product to produce in order to maximize profit for the quarter.

There are a number of limits on what SilComputers can produce. The major constraints are as follows:
1. Each computer (either notebook or desktop) requires a Processing Chip. Due to tightness in the market, our supplier has allocated 10,000 such chips to us.
2. Each computer requires memory. Memory comes in 16MB chip sets. A notebook computer has 16MB memory installed (so needs 1 chip set) while a desktop computer has 32MB (so requires 2 chip sets). We received a great deal on chip sets, so have a stock of 15,000 chip sets to use over the next quarter.
3. Each computer requires assembly time. Due to tight tolerances, a notebook computer takes more time to assemble: 4 minutes versus 3 minutes for a desktop. There are 25,000 minutes of assembly time available in the next quarter.

Given current market conditions, material cost, and our production system, each notebook computer produced generates $750 profit, and each desktop produces $1000 profit.

There are many questions SilComputer might ask. The most obvious are such things as “How many of each type computer should SilComputer produce in the next quarter?" “What is the maximum profit SilComputer can make?" Less obvious, but perhaps of more managerial interest are “How much should SilComputer be willing to pay for an extra memory chip set?" “What is the efect of losing 1,000 minutes of assembly time due to an unexpected machine failure?" “How much profit would we need to make on a 32MB notebook computer to justify its production?"

Linear programming gives us a mechanism for answering all of these questions quickly and easily. There are three steps in applying linear programming: modeling, solving, and interpreting.

2. A production planning problem

A factory makes 3 components, A, B and C using the same production process for each. A unit of A take 1 hr, a unit of B takes 0.75 hrs and a unit of C takes 0.5 hrs. In addition, C has to be hand finished, an activity taking 0.25 hrs per unit. Each week total production time (excluding hand finishing) must not exceed 300 hrs and hand finishing must not exceed 45 hrs.

The components are finally assembled to make two finished products. One product consists of 1 unit of A and 1 unit of C selling for 30 pounds whilst the other consists of 2 units of B and 1 unit of C and sells for 45 pounds. At most 130 of the first product and 100 of the second product can be sold each week. Formulate the problem of planning weekly production to maximize total profit and provide the optimal solution

3. A Market Allocation problem

MegaMarketing is planning a concentrated one week advertising campaign for their new CutsEverything SuperKnife. The ads have been designed and produced and now they wish to determine how much money to spend in each advertising outlet. In reality, they have hundreds of possible outlets to choose from. We will illustrate their problem with two outlets: Prime-time TV, and newsmagazines.

The problem of optimally spending advertising dollars can be formulated in many ways. For instance, given a fixed budget, the goal might be to maximize the number of target customers reached (a target customer is a customer with a reasonable chance of purchasing the product).

An alternative approach, which we adopt here, is to define targets for reaching each market segment and to minimize the money spent to reach those targets. For this product, the target segments are Teenage Boys, Affluent Women (ages 40-49), and Retired Men. Each minute of primetime TV and page of newsmagazine advertisement reaches the following number of people (in millions):

Outlet	Boys	Women	Men	Cost
TV	5	1	3	600
Mag	2	6	3	500
Target	24	18	24	

Again, MegaMarketing is interested in straightforward answers like how many units of each outlet to purchase to meet the segment goals. They are also interested in such questions as “How much will it cost to reach an extra million retired men?", “One radio spot reaches 1 million boys, 1 million women, and 1 million men: how much are we willing to pay for such a spot?", and similar questions.

4. Workforce Planning

Consider a restaurant that is open seven days a week. Based on past experience, the number of workers needed on a particular day is given as follows:

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Number	14	13	15	16	19	18	11

Every worker works five consecutive days, and then takes two days off, repeating this pattern indefinitely. How can we minimize the number of workers that staff the restaurant?

5. Financial Portfolio

In your finance courses, you will learn a number of techniques for creating “optimal" portfolios. The optimality of a portfolio depends heavily on the model used for defining risk and other aspects of financial instruments. Here is a particularly simple model that is amenable to linear programming techniques.
Consider a mortgage team with $100,000,000 to finance various investments. There are five categories of loans, each with an associated return and risk (1-10, 1 best):

Loan/investment	Return (%)	Risk
First Mortgages	9	3
Second Mortgages	12	6
Personal Loans	15	8
Commercial Loans	8	2
Government Securities	6	1

Any uninvested money goes into a savings account with no risk and 3% return. The goal for the mortgage team is to allocate the money to the categories so as to:
(a) Maximize the average return per dollar
(b) Have an average risk of no more than 5 (all averages and fractions taken over the invested money (not over the saving account)).
(c) Invest at least 20% in commercial loans
(d) The amount in second mortgages and personal loans combined should be no higher than the amount in first mortgages.

6. Production over Time

Many problems are multiperiod, where a series of decisions have to be made over time. For instance, in our very first SilComputer example, production decisions for computers have to be made with the future in mind. In this example, it would be easy to work with both desktop and notebook computers but let's work just with notebook computers.
SilComputer needs to meet the demand of its largest corporate and educational customers for notebook computers over the next four quarters (before its current model becomes obsolete).
SilComputer currently has 5,000 notebook computers in inventory. Expected demand over the next four quarters for its notebook is 7,000; 15,000; 10,000; and 8,000. SilComputer has sufficient capacity and material to produce up to 10,000 computers in each quarter at a cost of $2000 per notebook. By using overtime, up to an additional 2,500 computers can be produced at a cost of $2200 each. Computers produced in a quarter can be used either to meet that quarter's demand, or be held in inventory for use later. Each computer in inventory is charged $100 to reflect carrying
costs.

How should SilComputer meet its demand for notebooks at minimum cost?

7. A Distribution problem

A manufacturing firm produces widgets and distributes them to five wholesalers at a fixed delivered price of $2.50 per unit. Sales forecasts indicate that monthly deliveries
will be 2700, 2700, 9000, 4500 and 3600 widgets to wholesalers 1-5 respectively. The monthly production capacities are 4500, 9000 and 11,250 at plants 1, 2 and 3, respectively. The direct costs of producing each widget are $2 at plant 1, $1 at plant 2 and $1.80 at plant 3.
The transport cost of shipping a widget from a plant to a wholesaler is given below.

Wholesaler	1	2	3	4	5
Plant 1	.05	.07	.11	.15	.16
Plant 2	.08	.06	.10	.12	.15
Plant 3	.10	.09	.09	.10	.16

Formulate an LP model for this production and distribution problem.

8. The mining firm case study

A mining company produces and distributes the potassium hydroxide to various chemical customers localized in 4 regions: R1, R2, R3, R4. The product is sold under two configurations: powder and granule.

There are 2 manufacturing sites: the first one is localized in Gabon and the second one in Alsace (France). The Gabon site produces only the powder configuration which is supplied by boat (of capacity 10.000 tons) and then could be stored in the harbor warehouses. The site of Alsace supplies the warehouses of Havre, Saint Nazaire and Lyon, by complete trains (500 tons). The warehouses of Havre and Saint-Nazaire are equipped to transform some powder into granules - cost of the operation: 5 €/t All the customers are delivered from the stocks of the different warehouses. The transshipment cost is 3 €/t when the supply is done by train and 4 €/t when done by boat. The transportation costs are provided in the appendix We try to define a production and a distribution plane for the month to come which reduces the global cost of supplying the different customers (production + transportation). We consider the following commercial and technical constraints: (The unit used in the table is: 1 000 tons)

The estimated Demand per customer :

	R1	R2	R3	R4
Powder	150	50	100	70
Granules	50	50	50	30

The production capacity:

		Gabon	Alsace
Powder	Max	350	400
	Min	0	100
Granules	Max	0	200
	Min	0	0

The delivery capacity for each warehouse (all configurations included) and the inventory levels in the beginning of the month :

	Le Havre	Saint Nazaire	Lyon
Max delivery	300	300	200
Powder stock	50	100	0
Granules stock	0	0	0

Appendix
The unit holding ratio: 12% per year.
Chartering by boat: 80 k€ per travel (10 000 t)
Production costs:

GABON : 80 €/ t
ALSACE : 85 €/ t for the powder and 89 €/ t for the granules.

Supply side transportation costs, departure from Alsace to : Saint Nazaire : 3 €/ t Le Havre : 2,5 €/ t Lyon : 2 €/ t

Demand side transportation costs (€/ t) :

	R1	R2	R3	R4
Saint Nazaire	8	9	10	12
Le Havre	7	6	8	8
Lyon	9	8	6	7

9. Cee's Candy Company

Cee's Candy Company mixes its Rainbow Box from two basic confections – chocolates and pastels. To meet its packaging requirements and to reflect changing ingredient costs, the company runs a linear program periodically to determine the number of pounds of each type of candy to put into the mix. The costs and requirements for a one-pound box are provided on the right.

	Proportion of Candy Weight		Minimum Mix
Item	Chocolates	Pastels	Requirement
Nuts	0.15	0.05	0.10 lb
Soft Centers	0.50	1.00	0.60
Hard Centers	0.50	0	0.20
Cost per pound	$4.00	$1.00	

For example, in each pound of chocolate candy, 15% of the contents consist of nuts, 50% of the pieces have soft centers, and 50% have hard centers. These percentages do not add to 100% because nuts are included in some candies of both types (hard and soft centers). The total weight restriction guarantees that boxes contain at least 1 pound of candy. A batch of chocolate-coated candies is to be made using the same cream base. The chef wants to determine how many pounds to make of the creamy fudge and Burgundy cherry. Each pound of these candies requires the following amounts of ingredients:

Item	Flavor		Amount Available
	Creamy Fudge	Burgundy Cherry	
Pecan	0.5 oz	0.3 oz	500 oz
Chocolate	2.0 oz	2.0 oz	3,000 oz
Vanilla	3 ml	1 ml	2,400 ml
Cherries	0	24	28,000

All other ingredients are in plentiful supply. Each pound of Burgundy cherry contributes $1.00 to overhead and profit, while the creamy fudge brings $2.00. The total weight of the cherry candy must be at least half that of the fudge.

1. Consider the problem of deciding how much of each candy type to put into a Rainbow Box.

a) Formulate Cee's linear program and then solve it to determine the optimal weights of chocolates and pastels to put in a Rainbow Box. Assume that each box must contain exactly 1 pound. Suppose that the total weight restriction is relaxed, so that at least 1 pound of candy must be included in each box. Find the new most attractive corner and the optimal solution. (Minimizing cost, refer to the first table)

2. Formulate Cee's second linear program regarding the amount of the two candy types to make. Determine the optimal number of pounds to make of the two types. (Do not bother converting units of measurements, refer to the second table).